

E-commerce EDA Challenge - Answers

Phase 1 - Where to begin the audit

My first step is profiling: row and column counts, data types, then a per-column null count and a duplicate check. There are no repeated rows or order ids here. Most gaps are minor, roughly 3 percent across a handful of numeric fields, while return_reason is blank for any order that was never sent back. I also recompute the price fields by hand and find they line up perfectly, meaning final_price and total_amount are calculated from other columns rather than recorded separately.

Phase 2 - Is pricing skewed

It is. Measuring skewness gives roughly 2 for the price related fields and close to 4.4 for total_amount, and in every one of them the average sits well above the median, which is the classic sign of a long right tail. Taking the logarithm pulls those distributions back to almost symmetric, so the sensible preprocessing choice is a power or log transform rather than ordinary standard scaling.

Phase 3 - Categoricals that break the pipeline

loyalty_tier carries a natural ranking from Bronze up to Platinum, so encoding it as ordinal keeps that order whereas one-hot would lose it. sub_category has forty five distinct values, which makes one-hot expensive. city is misleading because its values do not belong to the listed country, for instance a Canada row showing Bangalore, so I leave it out. return_reason leaks the answer since it is only filled in after a return, and must be excluded. Everything else has few categories and is safe to one-hot.

Phase 4 - Why customers return products

Around twelve percent of orders come back, and the five stated reasons each account for roughly a fifth, with none dominating. When I break the return rate down by category, loyalty level or shipping time it barely moves from that twelve percent. The only faint trends are slightly higher returns on poorly rated items and on orders with very large discounts. Realistically the columns we have do not explain returns well.

Phase 5 - The ColumnTransformer pipeline

I route columns through a single transformer along four paths. Long tailed numeric fields are filled with the median, power transformed and then scaled. The remaining numeric fields are filled with the median and scaled. loyalty_tier passes through an ordinal encoder with its ranking specified. The low cardinality text columns are filled with the most frequent value and one-hot encoded, with unknown categories ignored so prediction never fails on a new value. Identifiers, the date, the two redundant price totals, city and return_reason are dropped, and the result comes out with no gaps remaining.

Phase 6 - Correlation and what predicts returns

Some inputs duplicate each other: product_base_price and final_price move together at 0.98, and total_amount is just base times quantity, so I removed final_price and total_amount. None of the fields actually track the return flag though; the highest is is_cancelled at about 0.09 and the rest are

essentially zero. A random forest does no better than guessing the majority outcome, so it effectively labels every order as kept. The takeaway is that a useful return model would need extra information such as past return behaviour or review content.